

RESEARCH ARTICLE

Semantic Identifiability under Band-Order Ambiguity in PlanetScope Landslide Segmentation

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ABSTRACT

Released HR-GLDD NumPy arrays do not state their channel order. The paper’s Red–Green–Blue–NIR listing and notebook display support RGBN more strongly, but missing array-construction records leave native PlanetScope BGRN as a conservative sensitivity. We rerun matched controls under both mappings and retain a contrast direction only when both agree. Index-specific F1 contrasts remain negative, while the index-modulated boundary effect changes sign. Generic boundary weighting remains positive in all 18 comparisons, with mapping-level means of $+0.96$ – $+1.13$ points. CAS and LRD do not confirm transfer. A publicly locked Sen12Landslides test passes only the NDVI-modulated-minus-plain-boundary margin, while all arm means are 1.21–3.14% F1. Sen12 provides no support for recurrence and does not refute the mechanism. HR-GLDD remains descriptive because event and coordinate identities are absent.

KEYWORDS

landslide segmentation; PlanetScope; band-order ambiguity; schema robustness; boundary weighting; mechanism attribution

Word count: 2,199 (Pandoc; abstract–conclusion, excluding tables, figures, displayed equations, references, declarations).

1. Introduction

High-resolution optical imagery enables detailed landslide mapping, but roads, river deposits, clearings, and rock share similar structure. HR-GLDD provides 3 m PlanetScope patches (Meena et al. 2023); CAS, LRD, and Sen12 supply external multi-sensor or named-band tests (Xu et al. 2024, Orynbaikyzy et al. 2025, Höhn et al. 2025).

NDVI is attractive because fresh failures can disrupt vegetation, but roads, agriculture, riverbanks, and shadows also create index edges. An index-modulated boundary loss changes both boundary emphasis and spectral modulation unless ordinary boundary weighting is controlled. Array semantics are equally load-bearing: assigning Red to the wrong column changes both NDVI and raw-edge controls.

Devara et al. compare empirical and U-Net Sentinel-2 extraction (Devara, Maurya, and Dwivedi 2024), while Wang et al. design edge enhancement (Wang, Li, and Shi

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2025). Such work motivates boundary outcomes and generic controls before vegetation-specific interpretation.

Known-order channel permutations test adaptability (Li et al. 2025), not unresolved release provenance. We apply established identified-set logic to an evidentially asymmetric two-mapping audit with controlled reruns; we do not claim a new sensitivity primitive or equal support for the mappings.

We therefore evaluate components and semantic assumptions rather than presuming a mechanism. Our contributions are:

- (1) a two-order sensitivity envelope that reports whether signed contrasts retain direction over the release-supported RGBN and conservative native-order BGRN mappings;
- (2) an executable audit that records RGBN’s stronger evidence, the absent array-construction provenance, and complete sensitivity runs;
- (3) capacity- and weight-matched zero, raw-edge, candidate-index, plain-boundary, and index-modulated controls;
- (4) outcomes selected without test-label threshold tuning, plus CAS, LRD, and publicly locked Sen12 stress analyses.

The envelope identifies negative index-versus-generic F1 directions and a positive generic boundary direction, but leaves the index-modulated boundary direction unidentified. None of three external analyses confirms transfer; their defects or degeneracy preclude robust rejection.

2. Related work

2.1. *Landslide segmentation*

U-Net and residual-block U-Net architectures provide standard segmentation baselines (Ronneberger, Fischer, and Brox 2015, Zhang, Liu, and Wang 2018). HR-GLDD is a PlanetScope benchmark (Meena et al. 2023), CAS provides multisensor RGB patches (Xu et al. 2024), and LRD and Sen12Landslides provide named Sentinel-2 data (Orynbaikyzy et al. 2025, Höhn et al. 2025). In Remote Sensing Letters, Devara et al. compare U-Net with an empirical Sentinel-2 rule (Devara, Maurya, and Dwivedi 2024). These performance studies do not isolate the active component.

2.2. *Spectral and edge guidance*

NDVI is a Red/NIR ratio, so a shared multiplicative term cancels algebraically; real mountain illumination still produces documented topographic sensitivity (Matsushita et al. 2007). TriGEFNet reports an NDVI-guided benefit (Zhang et al. 2026); spectral indices have also been combined with edge extraction (Zhang et al. 2023), and SAEENet adds generic boundary guidance (Wang, Li, and Shi 2025). Li et al. permute known HLS bands to test adaptability (Li et al. 2025); we instead audit unresolved released-array provenance with zero, raw, and equal-mass controls.

2.3. *Finite sensitivity and identified sets*

Functional ANOVA has shown that architecture, fine-tuning, learning strategy, and initialization contribute differently across remote-sensing classification datasets (Gho-

lami Shiri et al. 2026). That work decomposes variability across known configurations; here the uncertain input is physical channel identity. Multiverse analysis propagates reasonable coding choices (Steege et al. 2016), while partial identification retains sets under structural uncertainty (Manski, Sanstad, and DeCanio 2021). Qin and Tong make family-relative identified sets executable for evaluation claims (Qin and Tong 2026). Our min-max rule applies that established primitive to physical-schema uncertainty, with full retraining, same-class controls, and a hash-bound claim record.

2.4. Boundary weighting and evaluation

Pixel-wise boundary emphasis predates this application in U-Net (Ronneberger, Fischer, and Brox 2015). LSL-SS-Net uses a landslide-specific level-set loss, while E-MSLDNet reports edge-aware CAS segmentation with boundary metrics (Yang et al. 2024, Zhao et al. 2025). We do not propose another boundary architecture or loss; we test whether index-specific attribution survives plain-boundary same-class control and unresolved physical schema.

3. Controlled methods

3.1. Band semantics and spectral controls

Let $\mathbf{X} \in \mathbb{R}^{B \times 4 \times H \times W}$ denote the released HR-GLDD arrays. The paper’s band listing and notebook display support RGBN, but no construction code binds names to array indices. We therefore retain native-product BGRN as a lower-support sensitivity. Green/NIR occupy columns 1/3 in both; Red is column 0 under RGBN and 2 under BGRN. Define

$$V_o = \frac{X_{\text{NIR}} - X_{\text{Red}(o)}}{X_{\text{NIR}} + X_{\text{Red}(o)} + 10^{-6}}. \quad (1)$$

We compare three two-channel controls:

$$C_{\text{zero}} = (0, 0), \quad (2)$$

$$C_{\text{raw},o} = (|\nabla X_{\text{Red}(o)}|, |\nabla X_{\text{NIR}}|), \quad (3)$$

$$C_{\text{index},o} = (V_o, |\nabla V_o|). \quad (4)$$

The zero control preserves capacity without spectral input; the raw control tests whether optical edges explain any index advantage. Every output declares its candidate order.

3.2. Finite two-order sensitivity

For finite mapping set $\mathcal{O}$ and contrast $\delta(o)$, define

$$\mathcal{E}_\delta = \left[\min_{o \in \mathcal{O}} \delta(o), \max_{o \in \mathcal{O}} \delta(o) \right]. \quad (5)$$

Direction is retained only if $0 \notin \mathcal{E}_\delta$. Here $\mathcal{O} = \{\text{RGBN}, \text{BGRN}\}$. This finite sensitivity neither assigns equal plausibility nor provides a confidence interval.

3.3. Capacity-matched residual attention

All controlled-attention models use one residual encoder-decoder with channels (32, 64, 128, 256) and three skip gates. Resized controls join encoder features:

$$M_l = \sigma[\text{Conv}_{1 \times 1}(\text{ReLU}(\text{BN}(\text{Conv}_{3 \times 3}[F_l, C_l])))], \quad (6)$$

$$F_{l,\text{out}} = F_l \odot M_l + F_l. \quad (7)$$

Every control has 2,114,084 parameters; U-Net and ResU-Net are baselines.

3.4. Loss controls

For pixel i , the base map averages binary cross-entropy and focal loss (Lin et al. 2017) ($\alpha = 0.35$, exponent 2):

$$\ell_{\text{base},i} = \frac{1}{2}\ell_{\text{BCE},i} + \frac{1}{2}\ell_{\text{focal},i}. \quad (8)$$

Let $B_{\text{GT}} = \text{Dilate}_{3 \times 3}(Y) - \text{Erode}_{3 \times 3}(Y)$ be the one-pixel boundary band:

$$L_{\text{plain}} = \text{mean}_i\{\ell_{\text{base},i}[1 + \gamma B_{\text{GT},i}]\}, \quad (9)$$

$$L_{\text{index},o} = \text{mean}_i\{\ell_{\text{base},i}[1 + \gamma \widetilde{B_{\text{GT}}|\nabla V_o|_i}]\}, \quad (10)$$

where $\gamma = 2$. The tilde rescales each informative index-boundary map to the mean mass of B_{GT} and falls back to B_{GT} below 10^{-6} . Eight of 1,119 images fall back under each order; maximum mass mismatch is 2.98×10^{-8} . The comparison changes allocation, not mean added weight.

3.5. Evaluation quantities

We compute pooled/per-tile segmentation metrics and one-pixel-tolerance boundary F1. Operating points use the validation-score breakpoint nearest U-Net’s threshold-0.5 validation recall; test labels never select thresholds. Antialiased $128 \rightarrow 38 \rightarrow 128$ PlanetScope evaluation measures spatial-information loss, not Sentinel-2 radiometry.

4. Experimental design

4.1. Dataset and scope of inference

HR-GLDD provides 1,119/284/355 train/validation/test patches (Meena et al. 2023); positive-pixel prevalence is 10.09/10.17/10.92%. It supplies no row-level event, scene, or coordinates, so patch independence is unverified. The test set was previously inspected. Results are therefore corrective and descriptive across seeds, without tile-bootstrap inference or geographic-generalization claims.

4.2. Factorial controls

Zero, raw-edge, and candidate-index controls cross base and plain-boundary losses; one index arm uses equal-mass modulation. With U-Net and ResU-Net this gives nine configurations, each trained with seeds 42–44 under both mappings (54 runs). All use AdamW (Loshchilov and Hutter 2019), learning rate 1.5×10^{-3} , weight decay 10^{-4} , cosine decay, batch size 32, and 15 epochs; validation F1 selects the epoch.

4.3. External analyses

CAS compares base and plain-boundary loss post hoc without confirmation credit (Xu et al. 2024). LRD uses named Sentinel-2 bands and 28 EIDs, but lacks a third-party pre-access timestamp and has trigger-family overlap (Orynbaikyzy et al. 2025). Sen12Landslides freezes 21 checkpoints before authorized evaluation of 1,889 patches from ten inventories (Höhn et al. 2025); its five conditions cover two margins, generic-boundary recurrence, absolute performance, and spatial integrity. A disclosed SCL-255 correction changed no input, mask, endpoint, or condition. Supplementary Methods S1–S3 give all protocols and deviations.

4.4. Interpretation rules

Input specificity requires improvement over zero and raw controls under both orders. Loss specificity requires index modulation to beat equal-mass plain weighting with stable direction. Complete-system gains override neither control; LRD is ineligible for confirmation, and Sen12 requires all five registered conditions.

5. Results

5.1. Band-order sensitivity and matched controls

Table 1. Three-seed means under release-supported RGBN and conservative native-order BGRN sensitivity. BF1 uses one-pixel Chebyshev tolerance.

Configuration	RGBN F1	BGRN F1	RGBN BF1	BGRN BF1
U-Net/base	66.92	66.92	49.62	49.62
ResU-Net/base	68.55	68.55	52.93	52.93
Attn-zero/base	68.88	68.88	53.17	53.17
Attn-raw/base	69.05	69.13	52.92	53.15
Attn-index/base	69.04	68.92	52.76	52.83
Attn-zero/plain-boundary	69.99	69.99	56.04	56.04
Attn-raw/plain-boundary	69.88	70.19	55.85	56.15
Attn-index/plain-boundary	69.98	70.12	55.96	56.20
Attn-index/index-boundary	69.58	69.78	56.34	56.15

Table 1 reports every configuration under both candidate orders; Supplementary Table S4 gives detailed RGBN-candidate operating metrics. With base loss, candidate-index attention reaches 69.04% F1 under RGBN and 68.92% under BGRN. Raw candidate-Red/NIR edges reach 69.05% and 69.13%, while the capacity-matched zero control reaches 68.88% under both. Candidate-index input specificity is therefore unsupported regardless of order. All contrasts and envelopes below use unrounded run-level values; subtracting the table’s two-decimal cells can differ by 0.01 point.

Plain boundary weighting raises F1 in every seed and control mode under both orders. Mean F1 spans 69.88–70.19% and boundary F1 55.85–56.20%. Equal-mass index modulation changes F1 by -0.41 points under RGBN and -0.33 under BGRN. Its boundary-F1 contrast changes from $+0.38$ to -0.05 points, so spectral modulation is not order-robust. The order-stable result is generic boundary emphasis, not spectral identity.

5.2. Two-order sensitivity envelopes

Each envelope is the range of a contrast over both mappings. It records whether the sign survives this finite sensitivity, not statistical uncertainty. The index-input-versus-raw F1 envelope is $[-0.21, -0.01]$ points and the index-versus-plain-loss F1 envelope $[-0.41, -0.33]$, identifying negative directions over the two declared candidates without assigning equal plausibility. The corresponding boundary-F1 envelope $[-0.05, +0.38]$ crosses zero and is direction-unidentified. Generic plain-boundary F1 has envelope $[+0.96, +1.13]$; all 18 individual effects lie between $+0.42$ and $+1.62$ points. Supplementary Table S5 is the generated envelope summary.

5.3. External analyses

The post-hoc CAS sensitivity fails three of four criteria: plain boundary weighting changes mean region-macro F1 by -2.19 points and boundary F1 by -1.86 points, with mixed seed effects. It receives no confirmation credit (Supplementary Table S1).

The internally precommitted LRD run fails all four criteria but is not confirmation-grade. Validation F1 is 0.008–0.151; selected-threshold mean F1 changes by -4.73 points. Its frozen boundary effect is -24.10 points, but alternatives span -0.15 to -4.92 , and two protected EIDs share a trigger family with development. It is failed/indeterminate, not a robust rejection (Supplementary Method S2).

The publicly locked Sen12 test is verified-not-confirmed. Its sole passing condition is the NDVI-modulated-minus-plain-boundary F1 margin: -0.06 points ($[-0.41, +0.26]$); the registered upper endpoint is below $+1.00$ point. NDVI-attention minus raw-edge F1 is $+0.70$ points ($[+0.28, +1.16]$) and fails its margin. Generic boundary F1 is $+0.61$ ($[-0.09, +1.32]$), with 5/10 inventories positive. All arm means are only 1.21–3.14% F1. The registered transformed-envelope check also fails for one Kyrgyzstan pair, although native rectangles only share an edge. This provides no support for recurrence and no refutation (Supplementary Method and Table S3).

5.4. Operating point and spatial degradation

At threshold 0.5, boundary-weighted residual-attention FPR is 2.47–2.50%. Validation-matched recall is exact in 53/54 runs; combined matched FPR/precision spans 1.97–2.01%/78.41–78.74%. This does not support index-specific false-positive suppression.

Controlled $128 \rightarrow 38 \rightarrow 128$ degradation lowers F1 by 4.33–5.20 points across orders, identifying same-sensor spatial-information sensitivity rather than cross-sensor causation.

6. Discussion

6.1. *What the controls change*

RGBN has stronger release support, but absent construction provenance prevents treating its Red index as verified. BGRN is a conservative sensitivity, not an equally evidenced mapping. Index-versus-generic F1 directions survive this audit; the index-modulated boundary direction does not.

Generic boundary weighting improves all 18 HR-GLDD comparisons, but no external analysis confirms transfer. CAS is post hoc; LRD has weak validation, endpoint instability, and trigger overlap. Sen12 fails four of five conditions, including absolute performance. Its transformed-envelope integrity failure reflects native rectangles that only abut, not duplicated samples. Transfer is unsupported, not refuted.

This does not make NDVI universally unhelpful: prior work reports useful spectral processing and NDVI guidance (Devara, Maurya, and Dwivedi 2024, Zhang et al. 2026). Our result is diagnostic, not a competitive model claim.

6.2. *Interpretation under incomplete observability*

Band-ratio cancellation is exact only for shared multiplicative irradiance. Diffuse light, shadow, adjacency, reflectance, and atmosphere remain; an NDVI gradient also responds to any vegetation edge. Without illumination or background labels, the experiment cannot isolate scarps.

For incompletely observed remote-sensing systems, dataset-specific calibration is essential (Varotsos and Zhou 2026). We likewise separate same-sensor degradation from dataset shift and withhold event-level inference.

6.3. *Implications for benchmark practice*

Four recommendations follow: publish machine-readable channel names; bind transfer and uncertainty claims to event/scene/spatial units; include generic same-class controls (zero/raw-edge inputs and equal-mass plain boundary loss); and propagate unresolved schemas into a machine-readable envelope.

The compact HR-GLDD model is not a transferable operational system; parameter count alone says nothing about an emergency workflow.

7. Limitations

HR-GLDD lacks array-construction and row-level event/scene/coordinate provenance. Thus the finite two-mapping envelope neither identifies wavelengths nor supplies population uncertainty; prior test inspection makes evaluation corrective. Binary masks preclude category-specific error rates, and the 10-m control measures PlanetScope resampling rather than sensor transfer.

CAS changes several domains post hoc. LRD lacks an external timestamp, has trigger-family overlap, and is 84.1% empty with weak validation. Sen12 used development calibration; arm F1 is below 3.2%, and its conservative spatial bar fails for abutting tiles despite zero native positive-area overlap. These analyses permit neither confirmation nor refutation.

Three seeds measure optimization variation, not independent events. No mosaic, acquisition chain, payload, or responder workflow was evaluated; operational and geographic claims remain outside scope.

8. Conclusion

HR-GLDD lacks array-construction provenance binding band names to NumPy indices: RGBN has stronger release support, while BGRN is a conservative native-order sensitivity. The resulting envelopes give negative index-versus-generic F1 envelopes, a positive generic-boundary envelope, and a zero-crossing index-modulated boundary envelope.

Ordinary boundary weighting improves every HR-GLDD seed/control/order comparison. None of three external analyses confirms transfer: CAS is post hoc, LRD is indeterminate, and the publicly locked Sen12 test satisfies only one of five conditions with degenerate absolute F1. Generic boundary weighting is therefore dataset-local evidence, not a transferable prior; Sen12 provides no support for recurrence but does not establish refutation.

Machine-readable semantics, schema-propagated envelopes, same-class controls, independently timestamped protocols, and true independence units must precede physical or generalization interpretations.

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Data availability statement. The HR-GLDD arrays are at <https://doi.org/10.5281/zenodo.7189381>; LRD is at <https://doi.org/10.5281/zenodo.17007637>; and Sen12Landslides is at <https://huggingface.co/datasets/paulhoehn/Sen12Landslides>. Code, summaries, output hashes, and source/history assets are in <https://github.com/akssha74/S3Net-Landslide> (release v6.0.6). CAS imagery remains at <https://doi.org/10.5281/zenodo.10294997>; source terms bind.

References

- Devara, Meghanadh, Vipin Kumar Maurya, and Ramji Dwivedi. 2024. “Landslide extraction using a novel empirical method and binary semantic segmentation U-NET framework using Sentinel-2 imagery.” *Remote Sensing Letters* 15 (3): 326–338. <https://doi.org/10.1080/2150704X.2024.2320178>.
- Gholami Shiri, Maryam, Eva Tuba, Sašo Džeroski, Tome Eftimov, and Ana Nikolic. 2026. “Design Choices That Matter: A Functional ANOVA Analysis for Remote Sensing Multi-Label Classification.” *arXiv preprint arXiv:2608.04702* <https://doi.org/10.48550/arXiv.2608.04702>.

- Höhn, Paul, Konrad Heidler, Robert Behling, and Xiao Xiang Zhu. 2025. “A Spatio-Temporal Dataset for Satellite-Based Landslide Detection.” *Scientific Data* 12: 1772. <https://doi.org/10.1038/s41597-025-06167-2>.
- Li, Wenwen, Sizhe Wang, Hyunho Lee, Chenyan Lu, Sujit Roy, Rahul Ramachandran, and Chia-Yu Hsu. 2025. “Landslide Hazard Mapping with Geospatial Foundation Models: Geographical Generalizability, Data Scarcity, and Band Adaptability.” *arXiv* arXiv:2511.04474. <https://doi.org/10.48550/arXiv.2511.04474>.
- Lin, Tsung-Yi, Priya Goyal, Ross Girshick, Kaiming He, and Piotr Dollár. 2017. “Focal Loss for Dense Object Detection.” In *IEEE International Conference on Computer Vision (ICCV)*, 2999–3007.
- Loshchilov, Ilya, and Frank Hutter. 2019. “Decoupled Weight Decay Regularization.” *International Conference on Learning Representations* <https://doi.org/10.48550/arXiv.1711.05101>.
- Manski, Charles F., Alan H. Sanstad, and Stephen J. DeCanio. 2021. “Addressing Partial Identification in Climate Modeling and Policy Analysis.” *Proceedings of the National Academy of Sciences* 118 (15): e2022886118. <https://doi.org/10.1073/pnas.2022886118>.
- Matsushita, Bunkei, Wei Yang, Jin Chen, Yuyichi Onda, and Guoyu Qiu. 2007. “Sensitivity of the Enhanced Vegetation Index (EVI) and Normalized Difference Vegetation Index (NDVI) to Topographic Effects: A Case Study in High-Density Cypress Forest.” *Sensors* 7 (11): 2636–2651. <https://doi.org/10.3390/s7112636>.
- Meena, Sansar Raj, Lorenzo Nava, Kushanav Bhuyan, Silvia Puliero, Lucas Pedrosa Soares, Helen Cristina Dias, Mario Floris, and Filippo Catani. 2023. “HR-GLDD: a globally distributed dataset using generalized deep learning (DL) for rapid landslide mapping on high-resolution (HR) satellite imagery.” *Earth System Science Data* 15 (7): 3283–3298. <https://doi.org/10.5194/essd-15-3283-2023>.
- Orynbaikyzy, Aiyim, Felix Albrecht, Wei Yao, Mahdi Motagh, Wandi Wang, Sandro Martinis, and Simon Plank. 2025. “Landslide Mapping with Deep Learning: The Role of Pre-/Post-Event SAR Features and Multi-Sensor Data Fusion.” *GIScience & Remote Sensing* 62 (1): 2502214. <https://doi.org/10.1080/15481603.2025.2502214>.
- Qin, Xi, and Jizhou Tong. 2026. “What Does an Evaluation License? A Commit-Bound Census of Claim Replay in Inspect Evals.” *arXiv* arXiv:2608.19269. <https://doi.org/10.48550/arXiv.2608.19269>.
- Ronneberger, Olaf, Philipp Fischer, and Thomas Brox. 2015. “U-Net: Convolutional Networks for Biomedical Image Segmentation.” In *Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 234–241.
- Steege, Sara, Francis Tuerlinckx, Andrew Gelman, and Wolf Vanpaemel. 2016. “Increasing Transparency Through a Multiverse Analysis.” *Perspectives on Psychological Science* 11 (5): 702–712. <https://doi.org/10.1177/1745691616658637>.
- Varotsos, C. A., and G. Zhou. 2026. “Information-centred mathematical methods for microwave and LiDAR remote sensing of nonstationary environmental systems.” *Remote Sensing Letters* 17 (6): 659–672. <https://doi.org/10.1080/2150704X.2026.2661868>.
- Wang, Xin, Yingying Li, and Aiye Shi. 2025. “SAEENet: a semantic adaptive edge enhancement network for remote sensing image semantic segmentation.” *Remote Sensing Letters* 16 (10): 1109–1119. <https://doi.org/10.1080/2150704X.2025.2536240>.
- Xu, Yulin, Chaojun Ouyang, Qingsong Xu, Dongpo Wang, Bo Zhao, and Yutao Luo. 2024. “CAS Landslide Dataset: A Large-Scale and Multisensor Dataset for Deep Learning-Based Landslide Detection.” *Scientific Data* 11: 12. <https://doi.org/10.1038/s41597-023-02847-z>.

- Yang, Yueheng, Zelang Miao, Xiaojing Li, Hua Zhang, and Shuai Chen. 2024. “LSL-SS-Net: Level Set Loss-Guided Semantic Segmentation Networks for Landslide Extraction.” *GIScience & Remote Sensing* 61 (1): 2416727. <https://doi.org/10.1080/15481603.2024.2416727>.
- Zhang, Yue, Ruiqi Yang, Qinling Dai, Yili Zhao, Weiheng Xu, Jun Wang, and Leiguang Wang. 2023. “Boosting Semantic Segmentation of Remote Sensing Images by Introducing Edge Extraction Network and Spectral Indices.” *Remote Sensing* 15 (21): 5148. <https://doi.org/10.3390/rs15215148>.
- Zhang, Zhengxin, Qingjie Liu, and Yunhong Wang. 2018. “Road Extraction by Deep Residual U-Net.” *IEEE Geoscience and Remote Sensing Letters* 15 (5): 749–753. <https://doi.org/10.1109/LGRS.2018.2802944>.
- Zhang, Zirui, Qingfeng Hu, Haoran Fang, Wenkai Liu, Ruimin Feng, Shoukai Chen, Qifan Wu, Peng Wang, and Weiqiang Lu. 2026. “TriGEFNet: A Tri-Stream Multimodal Enhanced Fusion Network for Landslide Segmentation from Remote Sensing Imagery.” *Remote Sensing* 18 (2): 186. <https://doi.org/10.3390/rs18020186>.
- Zhao, Xingwang, Xingya Cao, Jingyu Yang, Guanzheng Zhao, Chao Liu, Chunyang Liu, and Yi Chang. 2025. “E-MSLDNet: A Dual-Enhanced Network with Edge-Aware and Multiscale Feature Fusion for Precision Landslide Detection in Complex Terrain.” *Measurement Science and Technology* 36 (9): 095403. <https://doi.org/10.1088/1361-6501/adffa3>.